

Observed Trends in the Tropical Pacific Zonal Sea Surface Temperature Gradient Are Statistically Significant Based on Multiple Null Hypotheses

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ABSTRACT: Climate conditions worldwide are influenced by the mean and variability of the tropical Pacific zonal sea surface temperature (SST) gradient. How this gradient responds to greenhouse gas forcing is, therefore, critical for accurate future climate projections. The nature of the response, however, remains debated: Historical model simulations favor a weakening trend, whereas observational records from the same period are characterized by a strengthening trend. To explain this model–observation discrepancy, some attribute the observed trend to internal variability or observational uncertainties, while others suggest that models may inaccurately simulate the radiatively forced response. Past studies have analyzed different trend intervals and observational datasets, potentially contributing to conflicting conclusions about whether observations reflect the forced response. We present a comprehensive analysis of observed zonal SST gradient trends and their statistical significance. We estimate observed trends over all 20-yr or longer intervals within the 1870–2024 period and subsequently evaluate these trends against a series of null hypotheses using bootstrapped ensembles of various statistical, conceptual, and geophysical models. Our analysis reveals that both strengthening and weakening trends are observed, depending on the analyzed intervals; however, intervals extending into the twenty-first century, particularly those since 1950 or those over a century or longer, exhibit statistically significant strengthening trends, suggesting that such trends are unlikely to have emerged from internal variability alone. This finding has implications for the historical and probable near-term transient responses, indicating they are likely radiatively forced. We confirm these findings with multiple observational datasets, demonstrating that data uncertainties minimally influence our conclusions.

SIGNIFICANCE STATEMENT: The response of the tropical Pacific Ocean to radiative forcing, such as greenhouse gas forcing, remains highly debated: Climate models' historical forcing experiments generally simulate a weakening trend in the west–east sea surface temperature gradient, while observations so far estimate a strengthening trend. Past studies, regardless of which response they supported, used different trend intervals and observational data. This study analyzes all multidecadal or longer trends in the observed gradient and their associated statistical significance across multiple datasets. Although both strengthening and weakening trends are present in observations, we find the emergence of statistically significant strengthening trends over multidecadal and centennial-scale intervals extending into the past several decades likely reflects the influence of historical radiative forcings.

KEYWORDS: Pacific Ocean; Tropics; Sea surface temperature; Climate change; Internal variability; Trends

1. Introduction

The tropical Pacific Ocean is characterized by warm sea surface temperatures (SSTs) in the west and cooler SSTs in the east. The difference between SSTs in these regions, referred to as the zonal SST gradient (hereafter denoted as $\Delta\text{SST}_{\text{W-E}}$), is maintained by the positive Bjerknes feedback (Bjerknes 1966, 1969). When $\Delta\text{SST}_{\text{W-E}}$ is strengthened, easterly trade winds intensify, which increases equatorial upwelling and the zonal

tilt of the thermocline, causing shoaling of the thermocline in the east. This shoaling further cools SSTs in the eastern Pacific cold tongue region, thus reinforcing $\Delta\text{SST}_{\text{W-E}}$ and the trade winds. Conversely, reduced $\Delta\text{SST}_{\text{W-E}}$ weakens the trade winds and upwelling, thus deepening the thermocline in the east and further reducing $\Delta\text{SST}_{\text{W-E}}$.

The tropical Pacific Ocean exerts a strong control over global and regional climates, as $\Delta\text{SST}_{\text{W-E}}$ and the associated spatial pattern of the tropical Pacific SSTs are intimately linked to the location of deep convection, which is then communicated via Rossby and Kelvin waves throughout the global atmosphere (Trenberth et al. 1998). For instance, the tropical Pacific Ocean is home to the leading mode of natural climate variability known as El Niño–Southern Oscillation (ENSO), in which the Bjerknes feedback and zonal and meridional exchanges of ocean heat content sustain swings between strengthened (La Niña) and weakened (El Niño) $\Delta\text{SST}_{\text{W-E}}$ phases on 2–7-yr time scales (Sarachik and Cane 2010). Shifts in the ENSO state lead to dry and wet anomalies around the Pacific basin by adjusting the strength and location of the tropical

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zonal atmospheric circulation, known as the Walker circulation (Taschetto et al. 2020). Associated changes in surface and atmospheric temperatures, as well as vertical wind shear, also impact tropical cyclone activity in the Pacific and Atlantic basins (Sobel et al. 2023). Moreover, the tropical Pacific Ocean is the largest oceanic source of natural carbon dioxide (CO_2) (Takahashi et al. 2009), and the variability in global CO_2 air–sea flux correlates with ENSO (e.g., McKinley et al. 2004; Rödenbeck et al. 2014; Wong et al. 2022). The Pacific decadal oscillation (PDO; Mantua et al. 1997) (defined as the leading pattern of North Pacific SST variability) also has strong expression on decadal-to-multidecadal time scales (Minobe 1999), including in the tropical Pacific SSTs (e.g., Newman et al. 2016; Zhang et al. 1997). PDO has hydroclimatic teleconnections similar to those for ENSO, including its well-studied connection to drought in the southwestern United States during the twenty-first century (e.g., Lehner et al. 2018; Seager et al. 2023; Jacobson and Seager 2025). Variations in the tropical Pacific Ocean may also have implications for transient, and perhaps equilibrium, climate sensitivity (Rugenstein et al. 2023).

Given these impacts, determining how $\Delta\text{SST}_{\text{W-E}}$ responds to external radiative forcing, particularly to greenhouse gas (GHG) forcing, is essential for accurate climate projections. The forced response of $\Delta\text{SST}_{\text{W-E}}$ has been widely investigated, but conflicting theories exist (Vecchi et al. 2008; Watanabe et al. 2024), with some arguing in favor of a strengthening $\Delta\text{SST}_{\text{W-E}}$ trend (e.g., Clement et al. 1996) and others proposing a weakening trend (e.g., Vecchi and Soden 2007) as a GHG-forced response. A prevailing theory in favor of a weakening gradient in response to increasing GHG forcing is that the Walker circulation weakens due to the nonlinear nature of the Clausius–Clapeyron relation: Increasing temperature with approximately constant relative humidity increases specific humidity at a faster rate than precipitation, requiring convective mass fluxes to reduce, thus weakening the Walker circulation (Vecchi and Soden 2007). Earlier work by Betts (1998) and Knutson and Manabe (1995) also argued that increased moist convection results in enhanced dry static stability, leading to a weakening of the Walker circulation. A weakened Walker circulation leads to reduced easterly trade winds that diminish $\Delta\text{SST}_{\text{W-E}}$ via the Bjerknes feedback. Based on the nonlinear Clausius–Clapeyron relation between changes in saturation vapor pressure and temperature, Knutson and Manabe (1995) additionally argued that latent heat loss due to evaporation requires less SST change in the western Pacific warm pool, relative to the eastern Pacific cold tongue, to balance radiative forcing; the net effect is to help dampen the SST increase in the western Pacific region, thus reducing $\Delta\text{SST}_{\text{W-E}}$.

In contrast to atmospheric arguments in favor of a weakening gradient, the ocean dynamical thermostat mechanism (Clement et al. 1996) proposes a strengthening of $\Delta\text{SST}_{\text{W-E}}$ in response to increasing atmospheric GHG concentrations. This mechanism hypothesizes that under such forcing, the tropical Pacific SSTs warm more in the west than in the east because equatorial upwelling and the shallow thermocline in the east allow warming to be offset by enhanced vertical advective cooling and dynamical ocean heat divergence. In the western Pacific warm pool, given the lack of upwelling and

deep thermocline, SSTs keep rising until the surface energy balance is restored through enhanced latent heat losses that balance reduced net upward longwave radiative heat fluxes [also see Sun and Liu (1996)].

The discussion around the response of the tropical Pacific SSTs to human influence is further complicated by the fact that observational records have shown $\Delta\text{SST}_{\text{W-E}}$ strengthening over the twentieth and twenty-first centuries, while the forced $\Delta\text{SST}_{\text{W-E}}$ change in historical model simulations favors weakening [see Lee et al. (2022) for a recent review]. This discrepancy between observations and climate models may arise from observational uncertainties, realizations of internal variability, or inaccuracies in the modeled response to historical radiative forcings. For instance, Coats and Karnauskas (2017) illustrated that the observed strengthening trend in $\Delta\text{SST}_{\text{W-E}}$ is not replicated at the same magnitude by any of the climate models participating in the Coupled Model Intercomparison Project phase 5 (CMIP5), arguing that model biases in the response to radiative forcing may contribute to the disparity between observations and climate models, as the models disagree on the underlying dynamical ocean mechanism even if they succeed in reproducing the strengthening trend. They also noted that this observation–model discrepancy is unlikely due to undersampling of internal variability, although they acknowledged that climate models may still underestimate internal variability [see also Laepple et al. (2023) and Sweeney et al. (2024)], thereby complicating the attribution process.

Observations limited to a single realization of internal variability also make it difficult and subject to considerable uncertainty to disentangle the components of the forced response from those of internal variability. To determine whether the observed trend is radiatively forced, past studies have evaluated the observed strengthening trend against the spread within climate model ensembles, which quantifies the range of modeled internal variability. Seager et al. (2019) reported that the observed trends in the Niño-3.4 SST index fall outside of the ensemble spread of historical model runs from CMIP5 and the National Center for Atmospheric Research (NCAR) Large Ensemble (LENS), arguing that the observed strengthening of $\Delta\text{SST}_{\text{W-E}}$ is indeed the forced response that the model simulations fail to capture. They subsequently employed a coupled atmosphere–ocean model of intermediate complexity to compare the CO_2 -forced $\Delta\text{SST}_{\text{W-E}}$ trends when the observed climatology is imposed, relative to those when the CMIP5 climatology is imposed. This work concluded that the biased climatology of state-of-the-art climate models (i.e., the cold tongue bias with relative humidity that is too high and wind speeds that are too weak) makes the SST in the cold tongue overly sensitive to radiative forcing.

Olonscheck et al. (2020) challenged the findings of Seager et al. (2019) and demonstrated that the observed $\Delta\text{SST}_{\text{W-E}}$ trends sit within the ensemble spread of CMIP5, CMIP6, and seven single-model ensembles, claiming that internal variability can explain the discrepancy between observed and simulated trends. They acknowledged, however, the possibility of systematic biases in modeled internal variability and forced responses. Supporting the claim by Olonscheck et al. (2020), Watanabe et al. (2021) also showed that the ensemble spread

of CMIP5, as well as four single-model ensembles, captures the observed $\Delta\text{SST}_{\text{W-E}}$ trend and suggested that a phase shift in a mode of internal variability, the interdecadal Pacific oscillation (IPO), may be related to the observed strengthening of $\Delta\text{SST}_{\text{W-E}}$. Conversely, Seager et al. (2022) confirmed the earlier findings of Seager et al. (2019) using CMIP6 and provided additional support for the influence of the ocean dynamical thermostat via greater thermocline shoaling in observations than in climate models, which they speculated helps explain why the observed $\Delta\text{SST}_{\text{W-E}}$ is more accentuated than the simulated one. Wills et al. (2022) analyzed 16 single-model ensembles and also demonstrated that the observed $\Delta\text{SST}_{\text{W-E}}$ strengthening falls outside the ensemble spread. They argued that the discrepancies between observations and model simulations are unlikely to be attributed solely to internal variability but rather are indicative of either a biased forced response, modeled internal decadal variability that is too weak, or a combination thereof. Approaching the debate from a different perspective, Heede and Fedorov (2023) decomposed spatial patterns in observational SST records and presented a tropical Pacific SST pattern that included Indo-west Pacific warming and relative cooling in the eastern tropical Pacific region that they argued is an anthropogenically forced transient response, which they speculated will soon reverse and lead to a weakening $\Delta\text{SST}_{\text{W-E}}$ following the delayed warming of subsurface waters.

All of the studies highlighted above used different intervals and observational data when analyzing $\Delta\text{SST}_{\text{W-E}}$ changes. Seager et al. (2019) focused on relatively long-term trends by examining ten 60-yr intervals ending between 2008 and 2017 using data from the Met Office Hadley Centre's Sea Ice and SST data set (HadISST), the U.S. National Oceanic and Atmospheric Administration (NOAA)'s Extended Reconstructed SST version 5 (ERSST.v5), the European Centre for Medium-Range Weather Forecasts (ECMWF)'s Ocean Reanalysis System 4 (ORAS4), and the NOAA National Centers for Environmental Prediction (NCEP)–NCAR Climate Data Assimilation System 1. One of those 60-yr intervals, the 1951–2010 interval, is also assessed in Watanabe et al. (2021) using HadISST, Hadley Centre SST version 3 (HadSST3), ERSST.v5, the Japan Meteorological Agency's Centennial in situ Observation-Based Estimates of SST version 2 (COBE2), KAPLAN from Kaplan et al. (1998), and the International Comprehensive Ocean–Atmosphere Data Set (ICOADS). Seager et al. (2022) used intervals, including the most extended ones, that begin any year between 1870 and 1957 and end in 2018 with data from HadISST, ERSST.v5, and the Japan Meteorological Agency's COBE and COBE2. They also closely analyzed the 1958–2018 trend using HadISST, ERSST.v5, COBE, COBE2, and ECMWF's ORAS5. Another long interval, the 114-yr-long interval from 1900 to 2013, was analyzed by Coats and Karnauskas (2017) using seven SST datasets, including HadISST, HadSST3, and COBE2. In contrast, Wills et al. (2022) based their findings on data from a shorter interval between 1979 and 2020 from ERSST.v5, COBE, and the Atmospheric Model Intercomparison Project SST boundary condition dataset. Similarly, Heede and Fedorov (2023) analyzed the 1980–2021

period using HadISST, ERSST.v5, COBE, and KAPLAN. Olonscheck et al. (2020) studied even shorter intervals (1915–44, 1945–74, 1975–2004, and 1995–2014) using ERSST.v5, COBE2, and the Hadley Centre's HadISST2. The range of the trend interval selections is illustrated in Fig. 1a, in which each interval examined in the aforementioned studies is depicted in a triangle plot that represents all 20-yr or longer trend intervals. Figures 1b–j further illustrate the impact of such differences, as the observed $\Delta\text{SST}_{\text{W-E}}$ linear trends may vary in terms of magnitude and even direction when different time intervals are chosen for analysis. Regardless of the specific intervals tested in each study, it is also clear from Fig. 1a that a large range of possible trend intervals remains untested by the highlighted work.

The potential impact of the differences between analysis intervals and observational records used in previous studies has not been evaluated, leaving open questions about the dependence of their conclusions on specific temporal intervals and datasets. Adapting the triangle plot from Rugenstein et al. (2023) that evaluated all intervals of 18 years or longer between 1950 and 2020, we specifically examine the dependence of $\Delta\text{SST}_{\text{W-E}}$ characterizations using an assessment of all possible interdecadal and centennial intervals between 1870 and 2024 over four SST datasets (HadISST, COBE, COBE2, and ERSST.v5). All of the observed trends are subsequently evaluated against a series of null hypotheses (statistical significance tests), including five different statistical noise models and two physically based simple models [a conceptual mathematical model based on Battisti and Hirst (1989) and the Zebiak–Cane intermediate complexity geophysical model]. Our collective analyses represent the most comprehensive evaluation of observed $\Delta\text{SST}_{\text{W-E}}$ trends and their associated significance. Based on the extensive significance testing that we perform, we argue that the response to historical radiative forcings, such as those from anthropogenic GHGs, cannot be ignored when explaining recent strengthening trends in observed $\Delta\text{SST}_{\text{W-E}}$ (note that, unless specifically stated, the term radiative forcing is used herein to collectively refer to all historical radiative forcing agents, such as anthropogenic GHGs, natural and anthropogenic aerosols, and stratospheric ozone).

2. Methods

a. SST data

We use four gridded observational SST datasets: HadISST (Rayner et al. 2003), COBE (Ishii et al. 2005), COBE2 (Hirahara et al. 2014), and ERSST.v5 (Huang et al. 2017). In each dataset, monthly average SST data are provided with gridded latitude–longitude resolutions of either $1^\circ \times 1^\circ$ (HadISST, COBE, and COBE2) or $2^\circ \times 2^\circ$ (ERSST.v5). In all datasets, temporal and spatial gaps in the sparser, earlier part of the data interval, are filled to provide complete spatiotemporal coverage. COBE is unique because it relies entirely on in situ SST measurements for its SST reconstruction. HadISST, COBE2, and ERSST.v5 employ in situ and satellite measurements (when available) to establish spatiotemporal patterns that are used to fill the data gaps. These patterns are then applied to both

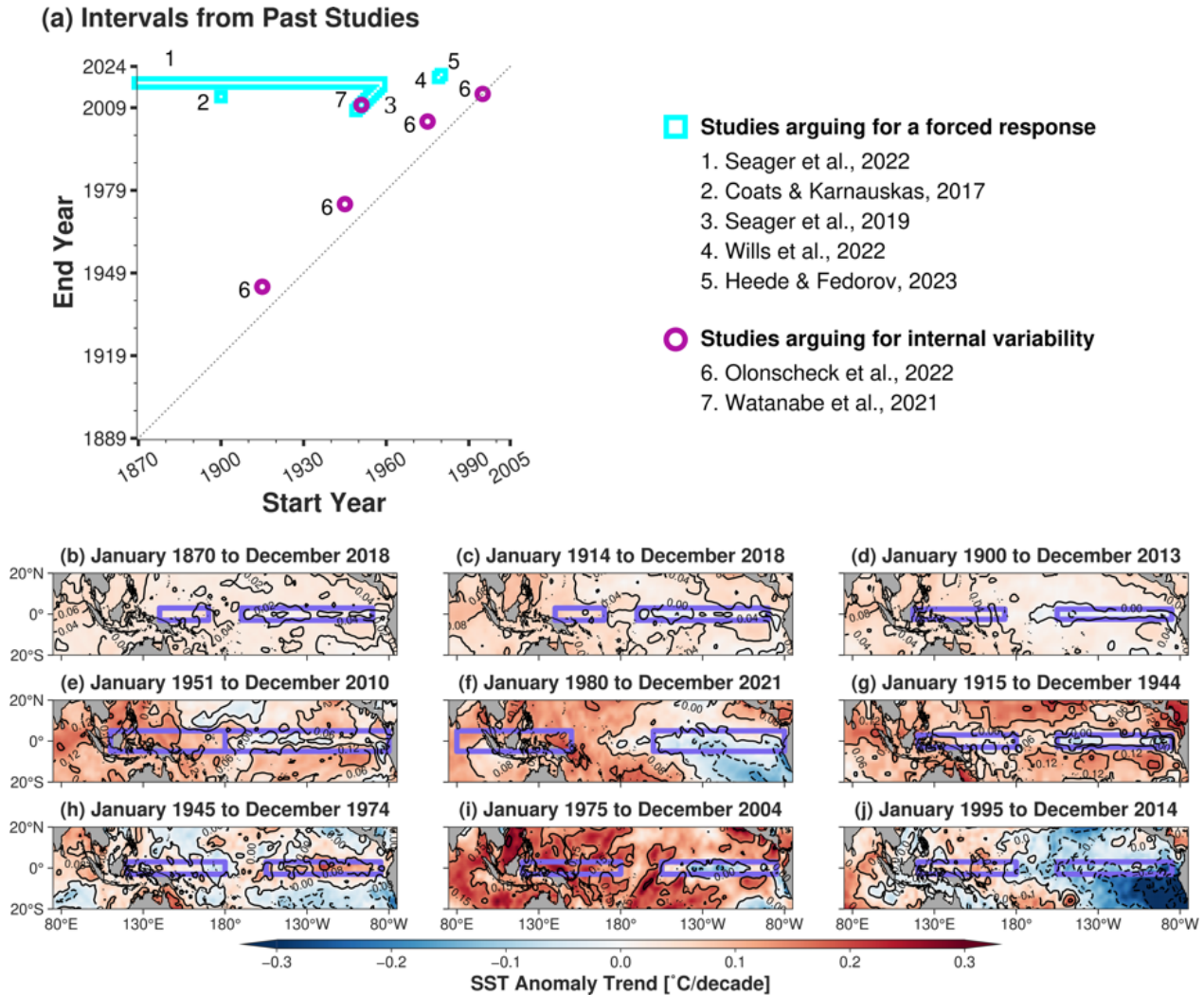


FIG. 1. (a) Triangle plot illustrating the intervals analyzed in previous studies. The x and y axes indicate the respective start (1870–2005) and end (1889–2024) years of the trend intervals. The minimum interval in the plot is 20 years, all of which are aligned along the diagonal axis (dotted line). Cyan boxes highlight the intervals analyzed in prior work that have concluded that historical radiative forcings have contributed to the observed $\Delta\text{SST}_{\text{W-E}}$ trend. Purple circles represent the intervals used by studies that have concluded that the observed $\Delta\text{SST}_{\text{W-E}}$ trend has arisen from internal variability alone. (b)–(j) Maps of the linear SST trend estimated from the HadISST data over interval(s) from (b), (c) Seager et al. (2022), (d) Coats and Karnauskas (2017), (e) Seager et al. (2019) and Watanabe et al. (2021), (f) Heede and Fedorov (2023), and (g)–(j) Olonscheck et al. (2020), with the purple boxes indicating the western and eastern equatorial Pacific regional box definitions used to compute $\Delta\text{SST}_{\text{W-E}}$ in each study (see section 2b for latitude and longitude ranges); note that in (e), only boxes from Watanabe et al. (2021) are shown. This study uses the same regional metrics as Seager et al. (2022) in (b) and (c).

in situ and satellite data in HadISST to construct complete SST fields, while COBE2 and ERSST.v5 only apply them to in situ data.

The data coverage of each dataset is as follows: HadISST starts from January 1870, COBE2 starts from January 1850, and ERSST.v5 starts from January 1854. COBE spans only January 1891–June 2023. Given these varying temporal coverages, we choose a shared interval from May 1870 to April 2025 for all datasets, except for COBE, for which we use May 1891–April 2023. Despite using all datasets, our primary focus is on HadISST in this study.

b. Computation of the observed trends in the zonal SST gradient

The $\Delta\text{SST}_{\text{W-E}}$ was defined by Seager et al. (2022) as the area-weighted mean SST in a western Pacific box (140° – 170°E , 3°N – 3°S) minus that in an eastern Pacific box (170° – 90°W , 3°N – 3°S) (purple boxes in Figs. 1b and 1c). Previous studies have proposed different west and east Pacific boxes: 117° – 173°E , 2.5°N – 2.5°S and 155° – 85°W , 2.5°N – 2.5°S in Coats and Karnauskas (2017) (Fig. 1d); 110° – 180°E , 5°N – 5°S and 180° – 80°W , 5°N – 5°S in Watanabe et al. (2021) [also used in Wills et al. (2022);

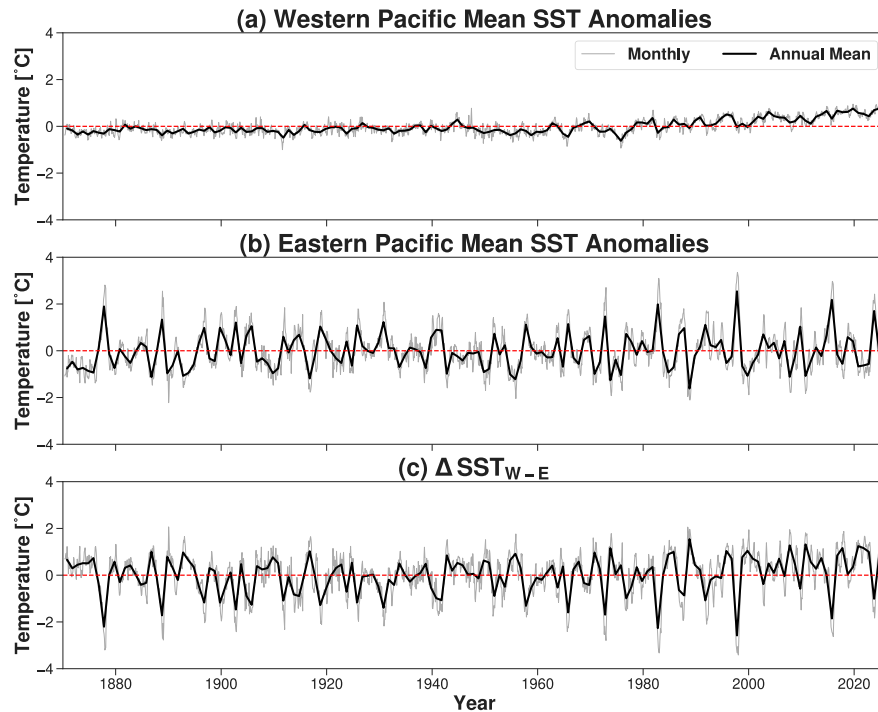


FIG. 2. The HadISST-derived monthly (gray) and annual (black) mean SST anomalies (a) in the western equatorial Pacific region (see the western purple box in Figs. 1b and 1c) and (b) in the eastern equatorial Pacific region (see eastern purple box in Figs. 1b and 1c). (c) The $\Delta S_{ST_{W-E}}$ time series, which is calculated by subtracting the mean SST anomalies of the eastern Pacific region from those of the western Pacific region (monthly in gray and annual in black). The horizontal line (dashed, red) in each panel indicates zero anomalies.

Fig. 1c]; 80° – 150° E, 5° N– 5° S and 160° – 80° W, 5° N– 5° S in Heede and Fedorov (2023) (Fig. 1f); and 120° E– 180° , 3° N– 3° S and 155° – 85° W, 3° N– 3° S in Olonscheck et al. (2020) (Figs. 1g–j). Despite these various definitions, Seager et al. (2022) argued that the lack of warming trend in the eastern Pacific is most evident in a narrow strip of the equatorial region where there is upwelling. Moreover, we have confirmed that, when computing the longest possible linear trend in the observed $\Delta S_{ST_{W-E}}$ time series for each dataset, the largest trend is obtained using the Seager et al. (2022) box selection in three out of four datasets (Table S1 in the online supplemental material). The influence of these variations in western and eastern Pacific regional metric definitions on estimates of observed $\Delta S_{ST_{W-E}}$ trends is nevertheless minor relative to the influence of trend interval choices and dataset selections (Fig. S1 in the online supplemental material). We thus base all subsequent analyses on the 3° N– 3° S box definition of Seager et al. (2022), as illustrated in Figs. 1b and 1c.

The $\Delta S_{ST_{W-E}}$ is calculated from monthly SST anomalies. From the original monthly mean SST time series, we first compute monthly anomaly time series for each grid using the climatology of a given dataset based on a common reference interval from May 1891 to April 2023. We subsequently take the area-weighted mean of the monthly SST anomaly time series for both the western (Fig. 2a) and eastern (Fig. 2b) boxes. A monthly $\Delta S_{ST_{W-E}}$ time series is then computed by subtracting the weighted area mean of the eastern Pacific SST anomalies

from that of the western Pacific SST anomalies (Fig. 2c). This monthly $\Delta S_{ST_{W-E}}$ time series is used to compute annual averages. In all cases, annual averages begin in May and end in April (e.g., the year 2000 is from May 2000 to April 2001) to preserve the effect of ENSO's seasonal evolution in SST anomalies (Rasmusson and Carpenter 1982).

The annually aggregated $\Delta S_{ST_{W-E}}$ time series spans 1870–2024 for HadISST, COBE2, and ERSST.v5 and 1891–2022 for COBE due to COBE's limited temporal coverage (see section 2a for details). Ordinary least squares regression is used to estimate a linear trend for every possible interval of 20 years or longer, with the longest trend spanning 132 years for COBE and 155 years for the other three datasets.

c. Null hypothesis tests

We propose a null hypothesis in which the observed $\Delta S_{ST_{W-E}}$ trends over all intervals that are 20 years or longer can be explained by one of several statistical noise models or physically based simple models that do not contain any secular forcing. Our base assumption is that if these noise or simple physical models encompass an observed trend over a given interval, those trends could be plausibly explained by internal variability as represented in the given null hypothesis model. Importantly, the variability in statistical or simple physical models is scaled to match the observational variability, which forms the basis of our interpretations of these models as

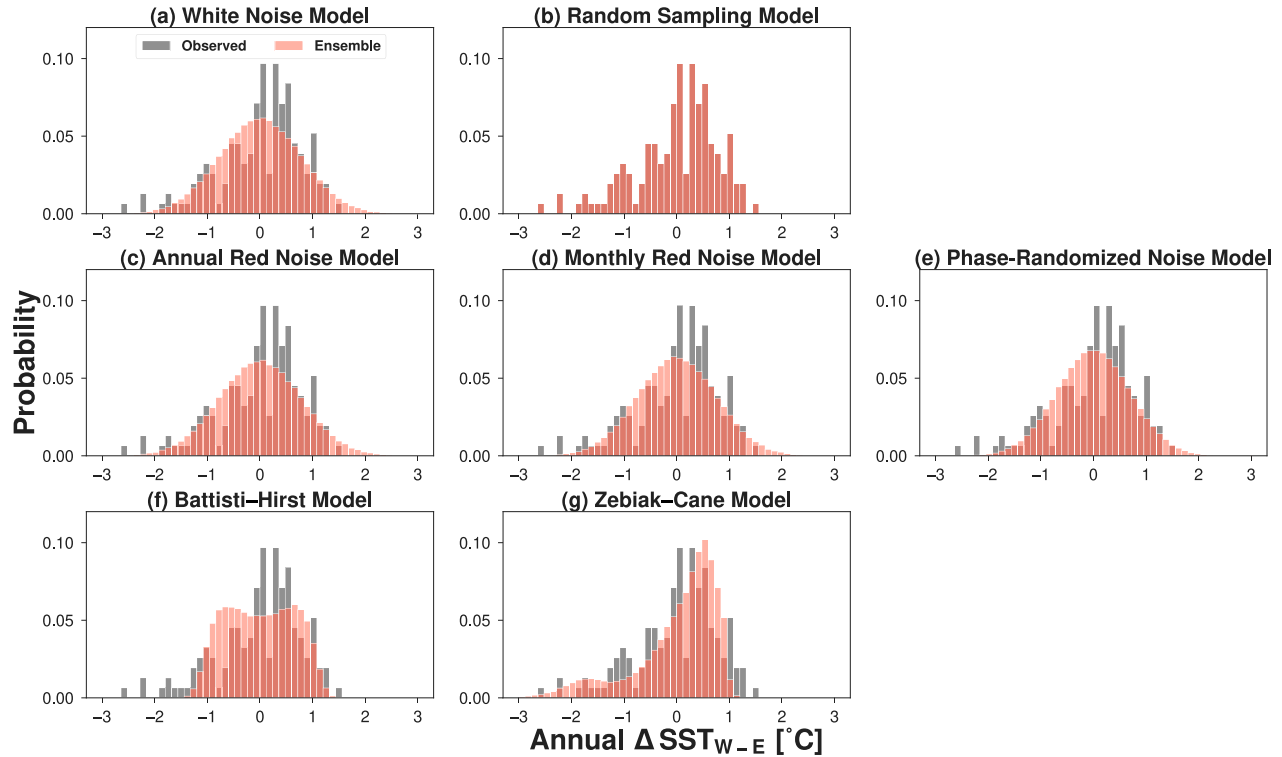


FIG. 3. The observed distribution (gray) of annual $\Delta\text{SST}_{\text{W-E}}$ values, compared with the ensemble distribution of (a) annual Gaussian white noise time series, (b) annual time series with randomly sampled observed annual $\Delta\text{SST}_{\text{W-E}}$ values, (c) annual red noise time series constructed using the observed annual AR1 coefficient, (d) monthly red noise time series constructed using the observed month-specific AR1 coefficients, (e) monthly time series with randomly shifted phases, (f) monthly time series constructed using the Battisti and Hirst (1989) conceptual model, and (g) monthly time series randomly sampled from a 100 000-yr unforced simulation of the Zebiak–Cane model. The ensemble distributions of all statistical and physical models are shown in red.

plausible estimates of internal variability (this approach is in contrast to other studies that have predominantly relied on characterizations of internal variability directly estimated by climate models). Conversely, we interpret trends that fall outside the range of those estimated by a given statistical or simple physical model as unexplained by internal variability and, therefore, potentially radiatively forced.

The five statistical and two simple physical models used in this study are described below, followed by additional details about the physics of SST variability they are intended to represent.

1) WHITE NOISE MODEL

Annual time series are derived from random draws from a normal (Gaussian white noise) distribution:

$$\Delta\text{SST}_{\text{W-E},n} = \sigma_{\text{obs}}\zeta + \mu_{\text{obs}}, \quad (1)$$

where $\Delta\text{SST}_{\text{W-E},n}$ is $\Delta\text{SST}_{\text{W-E}}$ at time $t = n$ and ζ is a random draw from a normal distribution $N(0, 1)$; μ_{obs} and σ_{obs} are the estimated mean and standard deviation from the observed annual $\Delta\text{SST}_{\text{W-E}}$ time series, respectively.

2) RANDOM SAMPLING MODEL

Annual time series are produced by randomly sampling (without replacement) the observed annual $\Delta\text{SST}_{\text{W-E}}$ values,

a process that is similar to white noise model construction but preserves the observed distribution of $\Delta\text{SST}_{\text{W-E}}$ values (note the observed and simulated distributions are fully identical in Fig. 3b).

3) ANNUAL RED NOISE MODEL

A lag-1 autoregressive (AR1) (red noise) model is given as

$$T_n = \alpha_Y T_{n-1} + \zeta, \quad (2)$$

where α_Y is the annual AR1 coefficient and T_n and T_{n-1} are the two consecutive time steps in the synthetic time series of $\Delta\text{SST}_{\text{W-E}}$. Conceptually, an AR1 coefficient, α , describes how much memory in $\Delta\text{SST}_{\text{W-E}}$ is carried from the previous time step, $t = n - 1$, into the contemporary time step, $t = n$. The larger the value of α , the greater the memory, or persistence, from the previous time step; if $\alpha = 0$, there is no memory in $\Delta\text{SST}_{\text{W-E}}$, and the synthetic time series is identical to the white noise model. Based on this annual red noise model, we construct annual time series using the value of α_Y estimated from the observed annual time series over the 1891–2022 interval (the period when all four datasets are available). The time series is rescaled to the mean (μ_{obs}) and standard deviation (σ_{obs}) of the observed annual $\Delta\text{SST}_{\text{W-E}}$ time series as follows:

$$\Delta\text{SST}_{\text{W-E},n} = \sigma_{\text{obs}}S(T)_n + \mu_{\text{obs}}, \quad (3)$$

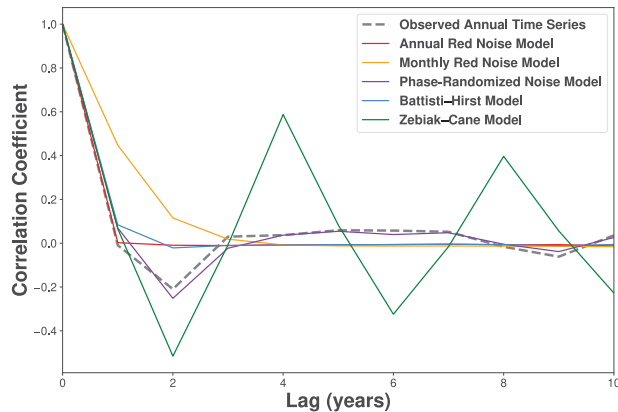


FIG. 4. Autocorrelation of the HadISST-observed $\Delta\text{SST}_{\text{W-E}}$ time series (dashed, gray), compared with the mean autocorrelation of 1000 time series employed in the null hypothesis tests: the annual red noise model (solid, red), the monthly red noise model (solid, orange), the phase-randomized noise model (solid, purple), the Battisti–Hirst model (solid, blue), and the Zebiak–Cane model (solid, green).

where the synthetically constructed time series, T , is centered and normalized by its mean and standard deviation and represented as $S(T)$. Note that, for several datasets, the observed α_Y value is close to zero (~ 0.0090 , ~ 0.0064 , ~ 0.059 , ~ 0.0085 estimated from HadISST, COBE, COBE2, ERSST.v5, respectively), making this hypothesis similar to the white noise test at the annual time scale. We highlight where this is the case in section 3, but we maintain the annual red noise test for completeness.

4) MONTHLY RED NOISE MODEL

This model is similar to the annual red noise model, but we use 12 distinct monthly AR1 coefficients (or α_M) estimated from monthly observations from May 1891 to April 2023 (again, the common temporal coverage of the four SST products) to generate synthetic monthly time series. We subsequently rescale this synthetic monthly time series to the mean (μ_{obs}) and standard deviation (σ_{obs}) of the observed monthly $\Delta\text{SST}_{\text{W-E}}$ time series after standardizing it, as shown in Eq. (3).

5) PHASE-RANDOMIZED NOISE MODEL

Monthly time series are derived by taking the Fourier transform of the linearly detrended observational data, randomly shifting the phases of the transformed data, and then back transforming the phase-shifted Fourier transform (Schreiber and Schmitz 2000). This method is a nonparametric approach to randomizing the data that approximately preserves the observed power spectral density and autocorrelation structure (see Fig. 4). The monthly time series is normalized and then scaled to the (nondetrended) observed monthly mean (μ_{obs}) and standard deviation (σ_{obs}) [see Eq. (3)].

6) BATTISTI–HIRST MODEL

This is a conceptual model proposed by Battisti and Hirst (1989) to describe the eastern Pacific mean SST variability as

$$T_n = T_{n-1} + bT_{n-1-\tau} + cT_{n-1} - eT_{n-1}^3 + \zeta. \quad (4)$$

Assuming no SST anomalies in the western Pacific region (as $\Delta\text{SST}_{\text{W-E}}$ is closely correlated with the eastern Pacific mean SST anomalies; approximately -0.94 in HadISST, -0.88 in COBE, -0.92 in COBE2, and -0.89 in ERSST.v5), we compute $\Delta\text{SST}_{\text{W-E}}$ based on this equation, employing the coefficient values of $b = 9.75 \times 10^{-2}$, $c = 1.83 \times 10^{-1}$, $e = 5.83 \times 10^{-3}$, and $\tau = 6$ months;¹ T_0 – T_6 and ζ are given by a random draw from a normal distribution $N(0, 2.5)$. The constructed monthly time series is then standardized and scaled to the mean (μ_{obs}) and standard deviation (σ_{obs}) of the observed monthly $\Delta\text{SST}_{\text{W-E}}$ time series, as seen in Eq. (3).

7) UNFORCED ZEBIAK–CANE MODEL

The Zebiak and Cane (1987) model is an intermediate complexity, coupled atmosphere–ocean model of the tropical Pacific Ocean. We construct a monthly time series by randomly selecting a 155-yr-long time series (or a 132-yr-long time series in the case of COBE; both time series begin in May and end in April) from a 100 000-yr unforced simulation performed by Ramesh and Cane (2019) and Ramesh et al. (2017). Each monthly time series is standardized and rescaled to the observed monthly mean (μ_{obs}) and standard deviation (σ_{obs}).

All models are used to derive bootstrapped ensembles of 1000 $\Delta\text{SST}_{\text{W-E}}$ time series with either monthly (for the monthly red noise model, the phase-randomized noise model, the Battisti–Hirst model, and the Zebiak–Cane model) or annual (for the white noise model, the random sampling model, and the annual red noise model) time steps. As noted in the model descriptions, every ensemble member (time series), except for the random sampling model, is then rescaled to have the same mean and standard deviation as the observed monthly $\Delta\text{SST}_{\text{W-E}}$ time series from May 1870 to April 2025 (or May 1891 to April 2023 for COBE) or annual time series over the 1870–2024 period (or 1891–2022 for COBE), depending on which time step is used for bootstrapping, as described in Eq. (3).

If an ensemble comprises monthly time series, each member is averaged into an annual May–April time series. From each member, a linear least squares $\Delta\text{SST}_{\text{W-E}}$ trend is calculated for all intervals ending in 2024 (or 2022 for COBE), with a minimum length of 20 years and a maximum length of 155 years (or 132 years for COBE), to establish ensemble distributions of $\Delta\text{SST}_{\text{W-E}}$ trends for each interval length. We evaluate the significance of the observed $\Delta\text{SST}_{\text{W-E}}$ trends against the ensemble distribution with the same interval length. A positive (strengthening) observed trend is considered significant if it is above the 95th percentile of that ensemble distribution; negative (weakening) trends are similarly evaluated against the 5th percentile.

All model ensembles attempt to mimic the range of internal variability arising from some collection of atmospheric and

¹ The coefficient b is modified from Battisti and Hirst (1989) in order to generate time series with a power spectral density that more closely matches the one from HadISST.

oceanic processes that produce SST anomalies in the tropical Pacific Ocean. The white noise and random sampling models are both white noise models, representing SST anomalies that emerge from stochastic atmospheric forcing (e.g., weather). The difference between the two white noise models is whether each time series is constructed based on a normal distribution (Fig. 3a) or constrained to have the same distribution as observations (Fig. 3b). However, these white noise models do not account for the lag between stochastic atmospheric forcing and SST anomalies as a result of air–sea heat fluxes and mixing within the ocean’s mixed layer (Frankignoul and Hasselmann 1977; Deser et al. 2010a); thus, we assume they do not capture all possible features of the observed $\Delta\text{SST}_{\text{W-E}}$ progression (although we note that the statistical significance reported in many of the previously mentioned studies is typically based on white noise null hypothesis assumptions). As an alternative statistical model, we explore red noise models (annually and monthly) that attempt to capture this lagged behavior, or the persistence, of SST anomalies by incorporating a lag-1 autocorrelation structure estimated from the observations. We then advance to constructing time series by random phase shifts (the phase-randomized noise model), which approximately preserve the full autocorrelation structure of the underlying observational data (i.e., it does not assume that the $\Delta\text{SST}_{\text{W-E}}$ variability is defined solely by an AR1 process). To complement the statistical null hypotheses, the conceptual Battisti–Hirst and the Zebiak–Cane geophysical models are adopted, as they explicitly take the growth of SST anomalies into account.

3. Results

a. Observed trends in the zonal SST gradient

Figure 5 plots the linear trends estimated from the observed $\Delta\text{SST}_{\text{W-E}}$ over all possible intervals of 20 years or longer between 1870 and 2024 (9316 intervals in total) and illustrates the presence of both strengthening (positive values; green grids) and weakening (negative values; purple grids) trends. Overall, there is a much larger number of intervals with a strengthening trend (75.8%), relative to those with a weakening trend (24.2%), and recent intervals overwhelmingly favor $\Delta\text{SST}_{\text{W-E}}$ strengthening; 99.0% of intervals ending in the last 20 years (2005 or later) exhibit a strengthening trend.

Along the diagonal axis, intervals with weakening trends alternate between intervals with strengthening trends, consistent with the pattern referred to as “swings” by Rugenstein et al. (2023). The most pronounced positive and negative trends emerge within those swings. If pronounced trends are characterized as those with a trend magnitude that is two standard deviations or more away from the mean of all the trends, roughly 73% of the pronounced strengthening trends sit over 20- to 30-yr intervals. The steepest strengthening trend is over the 20-yr interval from 1991 to 2010 ($0.495^{\circ}\text{C decade}^{-1}$). We note that pronounced weakening trends can have a similar trend magnitude; for example, the 1886–1905 period experienced the most intense weakening trend of $-0.571^{\circ}\text{C decade}^{-1}$. Approximately 80% of the pronounced weakening trends are with intervals of 20- to 30-yr lengths. This is not surprising

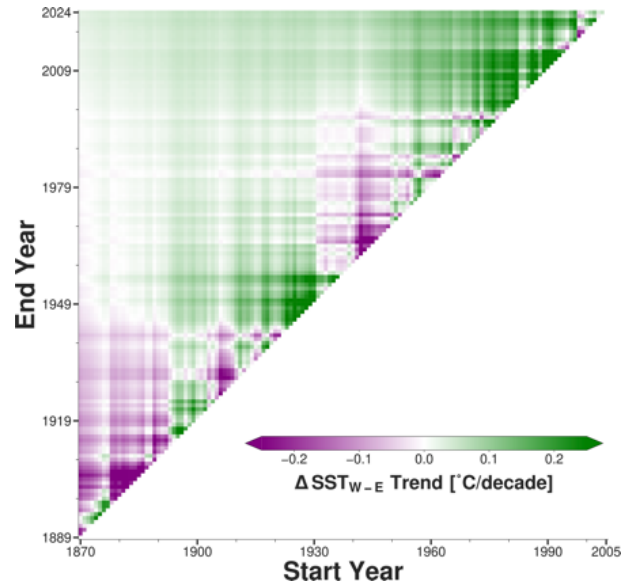


FIG. 5. Triangle plot showing the observed linear $\Delta\text{SST}_{\text{W-E}}$ trends in the HadISST dataset. Green and purple grids represent strengthening (positive) and weakening (negative) $\Delta\text{SST}_{\text{W-E}}$ trends, respectively. The shortest trend intervals (20-yr) are along the diagonal axis of the plot, while the longest trends (155-yr maximum) are in the top-left corner of the plot.

because internal variability on interannual time scales, the period with the largest variance, has the most influence on trends over shorter periods, while trends over longer intervals will be less impacted by this time scale of variability. Long-term intervals, such as those of 100 years or more, are primarily represented by a strengthening trend, with 94.9% of these centennial-length intervals showing strengthening, compared to 57.5% in 20- to 30-yr intervals.

b. Statistical significance of the observed trends

Figure 6 illustrates the results of the null hypothesis tests applied across all the observed $\Delta\text{SST}_{\text{W-E}}$ trends shown in Fig. 5. For each of the null hypotheses tested, the corresponding grid in Fig. 6 is shaded if the observed trend is significant (i.e., the null hypothesis fails): green if the observed strengthening trend is above the 95th percentile and purple if the observed weakening trend is below the 5th percentile.

The possibility that observed $\Delta\text{SST}_{\text{W-E}}$ trends are attributable to internal variability arising from stochastic atmospheric forcing is tested using the white noise hypothesis test. Approximately 12%–16% of the observed trends are statistically significant against this hypothesis test (Fig. 6a). However, this test fails to accurately replicate the observed distribution of $\Delta\text{SST}_{\text{W-E}}$ values (Fig. 3a), yielding an approximately symmetric distribution. Because the observed left-skewed $\Delta\text{SST}_{\text{W-E}}$ distribution reflects the asymmetric nature of ENSO variability, we also test a random sampling model that modifies the temporal sequence of the observed $\Delta\text{SST}_{\text{W-E}}$ values (this removes the observed autocorrelation structure, or persistence, of $\Delta\text{SST}_{\text{W-E}}$), while exactly preserving the skewed distribution of

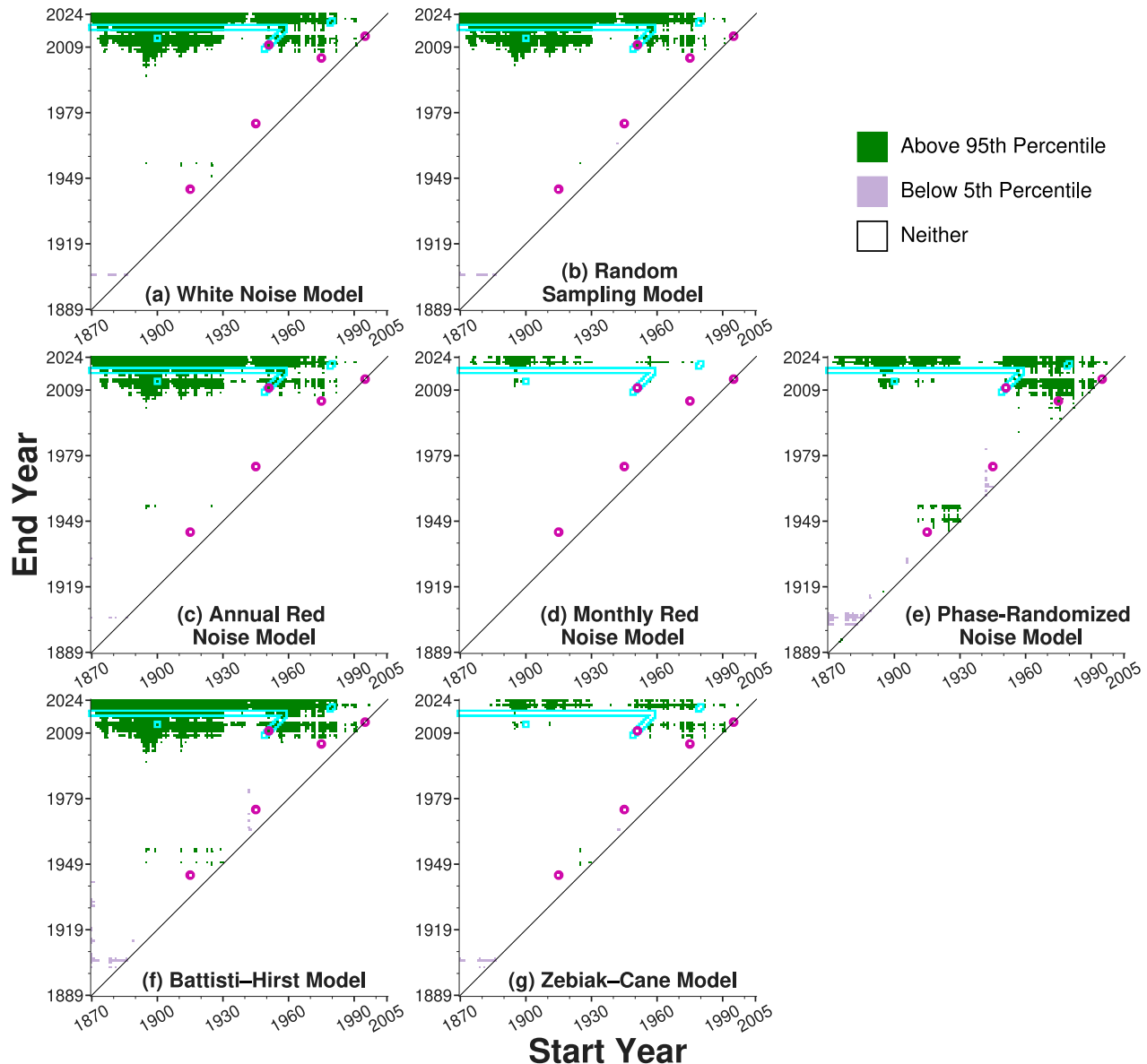


FIG. 6. Triangle plots demonstrating the significance of the observed ΔS_{T-W-E} trends from HadISST, tested against a bootstrapped ensemble of (a) annual Gaussian white noise time series, (b) annual time series with randomly sampled observed annual ΔS_{T-W-E} values, (c) annual red noise time series constructed using the observed annual AR1 coefficient, (d) monthly red noise time series constructed using the observed month-specific AR1 coefficients, (e) monthly time series with randomly shifted phases, (f) monthly time series constructed using the Battisti and Hirst (1989) conceptual model, and (g) monthly time series randomly sampled from a 100 000-yr unforced simulation of the Zebiak–Cane model. The significance is determined based on whether observed strengthening or weakening trends are above the 95th percentile (green) or below the 5th percentile (purple) of each ensemble distribution with the same interval length, respectively. Cyan boxes and purple circles indicate the intervals used by past studies, as shown in Fig. 1a.

the observations (Fig. 3b). Using this random sampling model as the null hypothesis yields a similar number of significant trend intervals ($\sim 13\%$ – 15% ; Fig. 6b) to those when using the white noise model. In both the white noise and random sampling hypotheses, the majority of the significant trends are with recent end dates [e.g., both significance tests indicate that more than 96% of the significant trends occur over intervals ending in the last 20 years (2005 or later) of our study period 1870–2024], and all of these trends indicate strengthening. Over half of these

recent significant strengthening trends are over intervals that are 100 years or longer. While some significant weakening trends are also present, they account for less than 2% of all significant trends and are mainly (i.e., more than 90% if not all) multidecadal trends ending more than five decades ago (before 1975).

Red noise processes are widely assumed to represent the lagged response of the ocean to stochastic atmospheric forcing via surface heat fluxes and mixing of heat within the

mixed layer, both of which create persistence (Frankignoul and Hasselmann 1977; Deser et al. 2010a). Such persistence is weakly manifest in the observed $\Delta\text{SST}_{\text{W-E}}$ time series at May–April annual time scales, with some datasets estimating AR1 coefficients that are close to zero (e.g., ~ 0.0090 for HadISST). We nevertheless include an annual red noise model for completeness, in addition to the fact that all four SST datasets estimate positive AR1 coefficients. Because the estimated AR1 coefficient in HadISST is close to zero, the results of the associated significance test are very similar to the two white noise estimates. Approximately 12%–14% of the observed $\Delta\text{SST}_{\text{W-E}}$ trends are significant using the annual red noise null hypothesis test, with the significant trends primarily associated with recent intervals (e.g., approximately 98%–99% of the significant trends have an end year in 2005 or later) (Fig. 6c). Similar to when tested against white noise tests, all of these recent significant trends exhibit strengthening, and more than half of them have 100-yr or longer intervals. We note that $\sim 1\%$ or less of the significant trends show weakening, but they are mostly (i.e., more than 90% if not all) limited to multidecadal trends with end dates before 1975.

ENSO has a characteristic seasonal evolution with anomalies that grow in the boreal summer and fall and peak in amplitude in the following winter (Rasmusson and Carpenter 1982; Torrence and Webster 1998; Zebiak and Cane 1987), which cannot be captured by the annual red noise model. This motivates the formulation of a red noise model using distinct monthly AR1 coefficients determined from observations to capture the seasonal cycle of SST anomalies and their persistence. The proportion of trends that are considered significant is substantially reduced when tested against the monthly red noise model ($\sim 3.5\%$ or less; Fig. 6d), relative to the annual red noise model, making this statistical noise model the most conservative (i.e., the hardest to reject the null hypothesis) among all seven tests explored in this study. Compared to the annual red noise model, the significant trends are further confined to even more recent intervals; all of the significant trends end in the last two decades. In fact, more than 85% of the significant trends are strengthening trends with an end date in the last 10 years (2015 or later) of our study period when tested using the monthly red noise model, compared to $\sim 63\%$ – 66% when tested using the annual red noise model. There are also no significant weakening $\Delta\text{SST}_{\text{W-E}}$ trends. Critically, however, the annual AR1 coefficient calculated from the time series generated with the monthly red noise model is markedly higher (e.g., ~ 0.45 for HadISST) than the observed value calculated with annual data (e.g., ~ 0.0090 for HadISST), suggesting that the monthly red noise model yields an annual persistence that is too large, and thus too conservative, relative to the observed temporal evolution of $\Delta\text{SST}_{\text{W-E}}$ (also see Fig. 4, showing that the autocorrelation of the monthly red noise model is consistently higher for 1- and 2-yr lags).

Neither the annual AR1 model nor the monthly AR1 model preserves the complete autocorrelation structure of observational data. The phase-randomized noise model, on the other hand, avoids the AR1 approximation and better preserves the complete autocorrelation structure of the observed $\Delta\text{SST}_{\text{W-E}}$ time series (Fig. 4). Tested against this phase-randomized noise

model, some multidecadal intervals now become significant, while the percentage of significant trends ($\sim 8\%$ – 11% ; Fig. 6e) falls between those tested against the annual ($\sim 12\%$ – 14%) and monthly ($\sim 3.5\%$ or less) red noise models. Strengthening trends with recent end dates (2005 or later) still dominate the significant trends (more than 80%). Notably, multidecadal periods starting in 1950 or later constitute roughly 45%–60% of these recent strengthening trends that are significant (compared to a maximum of $\sim 30\%$ in other null hypotheses mentioned so far). While some significant weakening trends ($\sim 6\%$ – 9% of the significant trends) are present, more than 97% of them are multidecadal with end dates before 1975.

Unlike the red noise models that only consider the persistence of SST anomalies on top of stochastic atmospheric forcing, the Battisti and Hirst (1989) model also takes the growth of SST anomalies into account. The null hypothesis test against this conceptual model suggests that $\sim 17\%$ – 19% of the intervals produce significant trends, mainly consisting of strengthening trends with recent end years (e.g., more than 90% of significant trends are strengthening trends over intervals ending in 2005 or later) (Fig. 6f). Roughly 50% of these recent significant strengthening trends are century-length trends or longer. In contrast, $\sim 3\%$ or less of the significant trends are accounted for by weakening trends, with over 90% of these significant weakening trends being multidecadal and ending before 1975.

These results are complemented by the test using the Zebiak–Cane model. Substantially fewer intervals (less than 5%) exhibit significant trends when tested against this model, making it the second most conservative model after the monthly red noise model (Fig. 6g). This result aligns with the fact that the bootstrapped ensemble constructed using the Zebiak–Cane model yields larger absolute autocorrelation with lags that are 2 years or longer (Fig. 4). The significant trends are still largely confined to intervals with recent end dates (e.g., more than 90% are with interval end dates in 2005 or later). All these significant recent trends indicate strengthening, and more than 90% of them are either over centennial-scale intervals or intervals since 1950. Less than 6% of the significant trends are weakening trends, though they do not occur over recent intervals (all significant weakening trends have multidecadal lengths with end dates prior to 1975).

c. Dataset comparison

To assess the robustness of the observed $\Delta\text{SST}_{\text{W-E}}$ trends, Fig. 7 compares observed $\Delta\text{SST}_{\text{W-E}}$ trends computed from HadISST, COBE, COBE2, and ERSST.v5. Overall, the characterization of the observed $\Delta\text{SST}_{\text{W-E}}$ trends between 1870 and 2024 (between 1891 and 2022 for COBE) is consistently similar across these four datasets; for instance, there are more strengthening than weakening trends across all four products. In COBE2 and ERSST.v5, strengthening trends account for 72.5% and 60.2% of the intervals, respectively. COBE has the highest percentage (84.7%) of strengthening trends, although the absence of SST data prior to 1891 and after 2022 is a contributing factor (if we only consider intervals from COBE's temporal coverage, the percentages of strengthening trends increase to 83.2% for HadISST, 80.7% for COBE2, and 70.5%

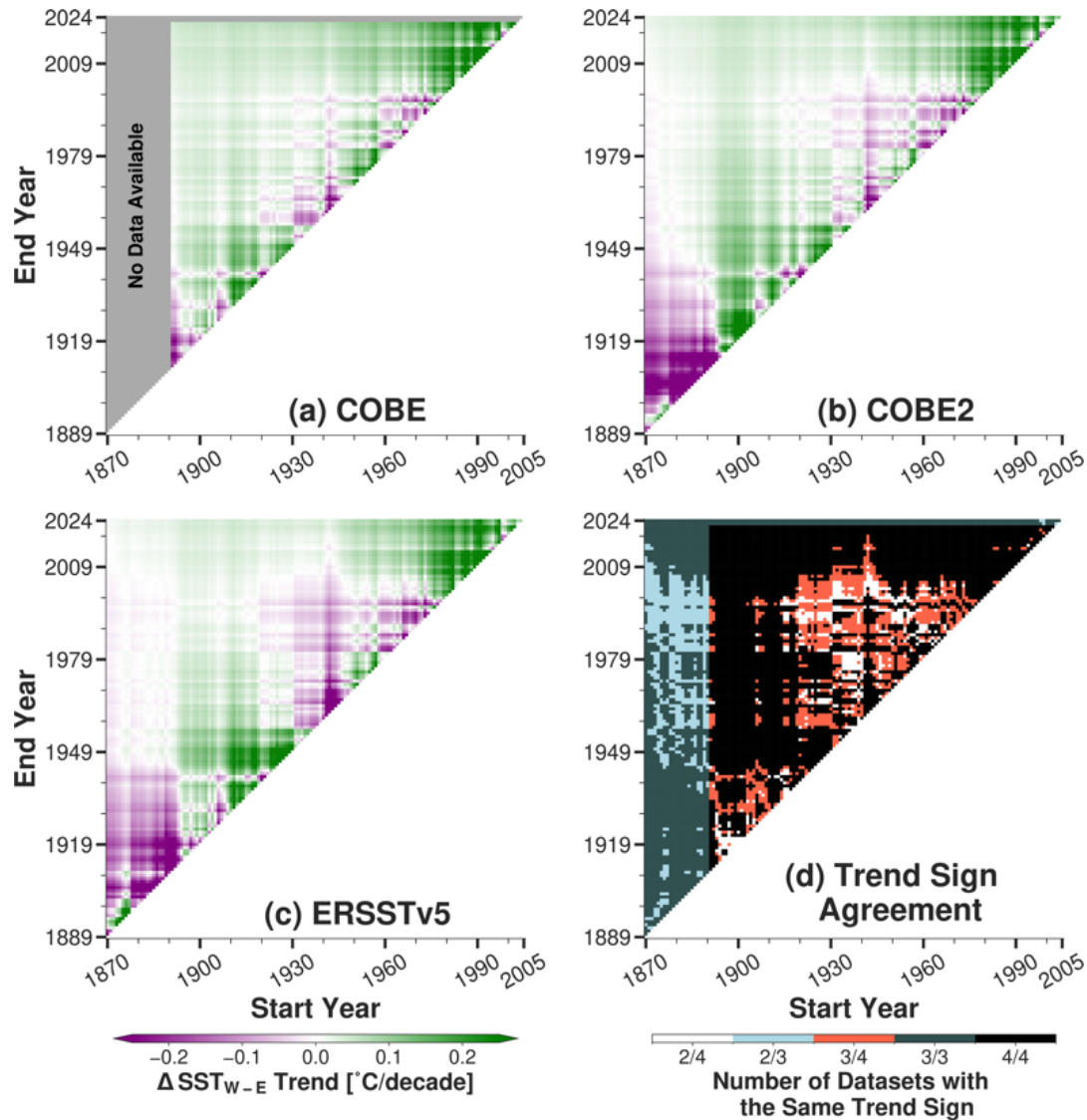


FIG. 7. Dataset comparison of observed $\Delta\text{SST}_{\text{W-E}}$ trends. Observed trends computed from (a) COBE, (b) COBE2, and (c) ERSST.v5 are presented. The color scale for (a)–(c) is as in Fig. 5. (d) The number of datasets (HadISST, COBE, COBE2, and ERSST.v5) that have the same trend sign in each interval grid (note that COBE is only available from 1891 to 2022).

for ERSST.v5, all of which are more comparable to COBE). Additionally, centennial-length intervals have a greater proportion of strengthening trends compared to shorter intervals of 20 to 30 years (100% vs 63.6% in COBE, 89.8% vs 57.5% in COBE2, and 68.3% vs 53.4% in ERSST.v5).

More importantly, 73.8% of intervals show a trend sign that is consistent across all datasets. Among the intervals with a trend-sign disagreement, 31.8% of intervals are from the period with only three datasets (i.e., intervals outside COBE's temporal coverage). For the remaining 68.2% of intervals with trend-sign disagreements, 32.2% of them have two datasets supporting each trend direction. In other words, maximal trend-sign disagreement (two data products support each trend sign) occurs over only 22.0% of all intervals with trend-sign

disagreement (or only 5.8% of all multidecadal and centennial intervals between 1870 and 2024). Moreover, intervals ending in the last two decades (2005 or later) are predominantly represented by a strengthening trend (99.3% in COBE, 99.1% in COBE2, and 94.0% in ERSST.v5), and only 6.0% of these intervals with end dates of 2005 or later have a trend-sign disagreement.

We also repeat all the null hypothesis tests using COBE, COBE2, and ERSST.v5 (Figs. 8–10). In general, ERSST.v5 yields the lowest percentage of intervals with significant trends evaluated against all null hypotheses. For each dataset, three of the hypothesis tests (based on the white noise, random sampling, and annual red noise models) yield a similar percentage of significant observed trends (~10%–13% for COBE,

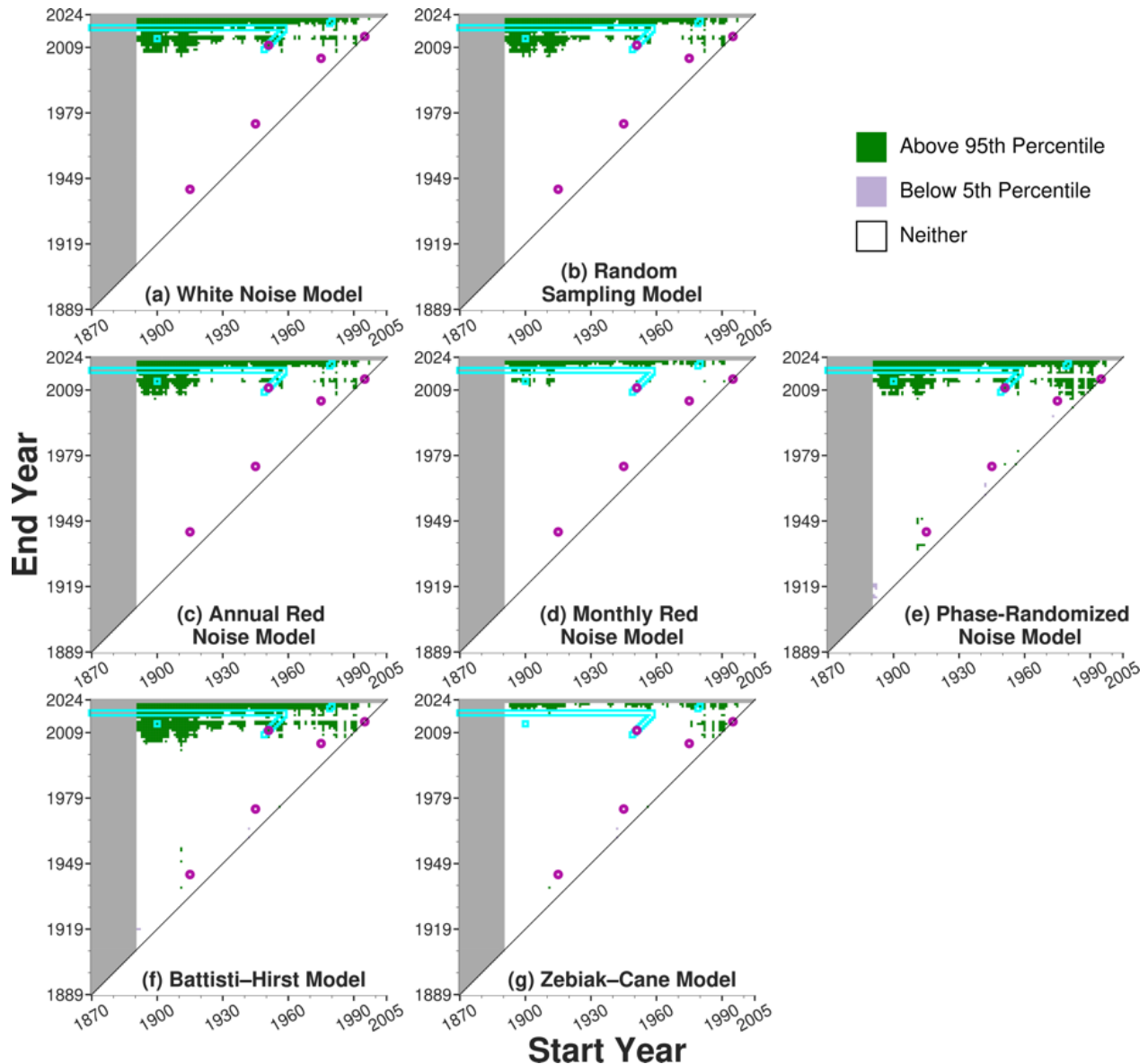


FIG. 8. As in Fig. 6, but using COBE data. The gray shading notes COBE's reduced temporal coverage relative to the other datasets.

~6%–12% for COBE2, ~2%–4% for ERSST.v5). The null hypothesis test based on the Battisti–Hirst model has the highest percentage of significant trends (~14%–17% in COBE, ~15%–18% in COBE2, and ~6%–8% in ERSST.v5). The lowest percentage of significant trends is recorded when tested against either the Zebiak–Cane model (less than 4% of all trends are significant in COBE) or the monthly red noise model (less than 1% in COBE2 and almost none in ERSST.v5).

Overall, the key findings are consistent across the four datasets: Significant trends are present over intervals with recent end dates, except for ERSST.v5 when assessed against the most conservative model—the monthly red noise model. Specifically, more than 95% of the significant trends in COBE occur over intervals ending in 2005 or later across all hypothesis tests; 40%–70% of the significant trends in COBE2, aside

from ~80% or more with the monthly red noise model; and 40%–55% of the significant trends in ERSST.v5, with exceptions for the monthly red noise (almost all trends are insignificant) and Zebiak–Cane (roughly 30%–40%) models (Figs. 8–10). These recent significant trends are all strengthening, mostly representing either multidecadal ones since 1950 or centennial ones (together accounting for ~69% or more in COBE, over 85% in COBE2, and 97%–100% in ERSST.v5). Although some null hypothesis tests for each dataset show some significant weakening trends, these are not recent trends, as their end dates are all over two decades ago (before 2005); in fact, roughly 90% or more of the significant weakening trends are multidecadal trends that end before 1975. Furthermore, regions of significant trends overlap consistently across datasets (Fig. 11). All null hypothesis tests other than the monthly red noise model

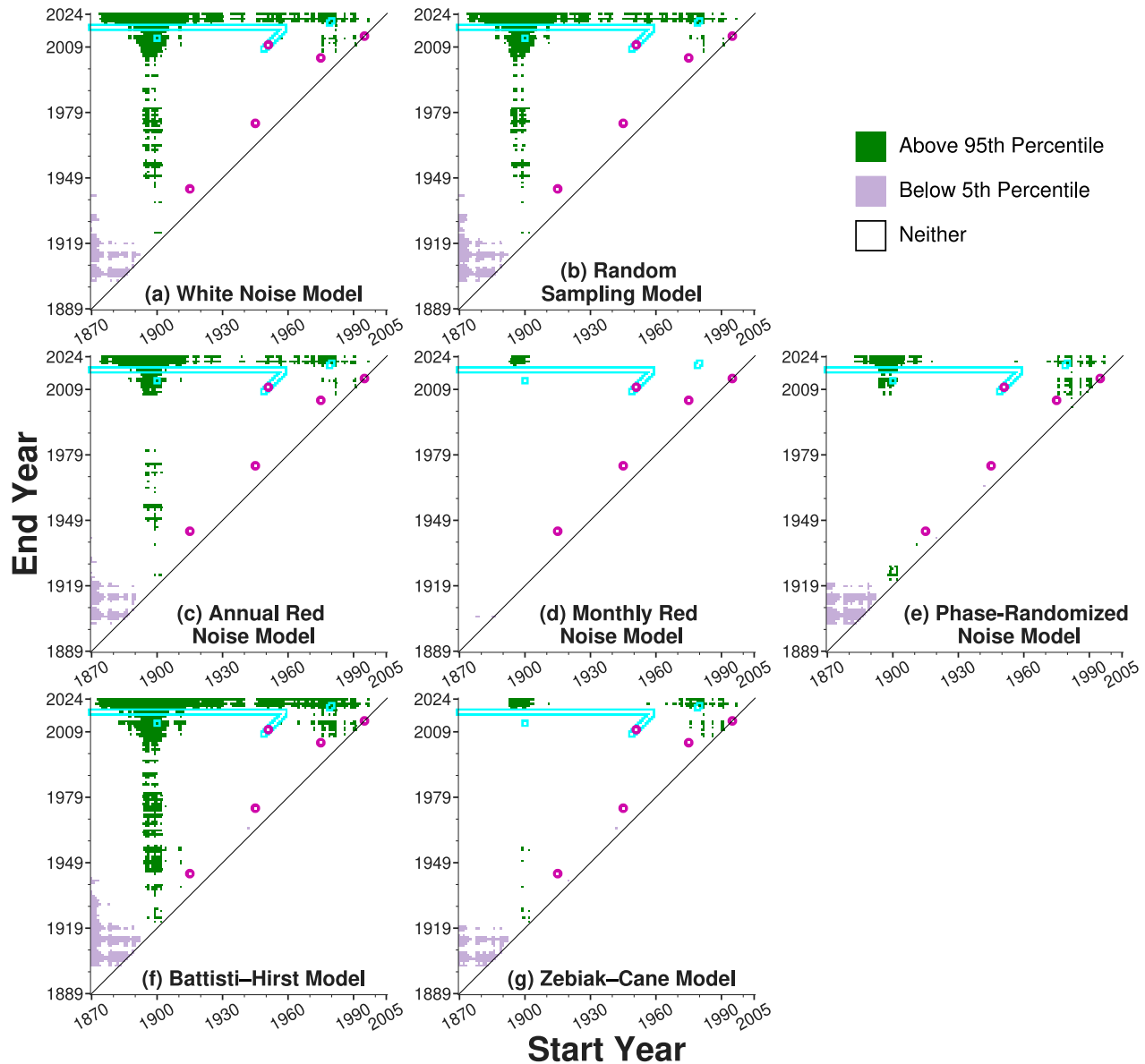


FIG. 9. As in Fig. 6, but using COBE2 data.

indicate that $\sim 75\%$ or more of intervals that are considered significant across all four datasets have end dates of 2005 or later, all exhibiting strengthening. These recent significant trends are predominantly ($\sim 97\%$ – 99% if not all) characterized as either centennial-scale trends or multidecadal-scale trends since 1950.

4. Discussion and conclusions

Our results demonstrate that observed $\Delta\text{SST}_{\text{W-E}}$ trends exhibit both strengthening and weakening periods, depending on interval choices; these results are consistent with the interdecadal alternating patterns that are referred to as swings of the $\Delta\text{SST}_{\text{W-E}}$ trends by Rugenstein et al. (2023) (Fig. 5). We also note that these swings are less prevalent as the time

period over which trends are evaluated becomes longer. For the longest intervals with a recent end date, these trends are predominantly strengthening. Additionally, our analysis suggests that significant trends, which we interpret herein as likely a radiatively forced, transient response because they are outside the range of statistically or physically estimated internal variability, include strengthening trends over intervals leading up to the last two decades of our study period. This is particularly true for multidecadal strengthening trends over intervals beginning in 1950 or later, as well as for centennial-scale strengthening trends, despite their smaller trend magnitudes. Significant weakening trends exist, but they mostly occur at the multidecadal scale over intervals that end before 1975. Because disagreements over the eastern equatorial Pacific SST trends among interpolated and uninterpolated

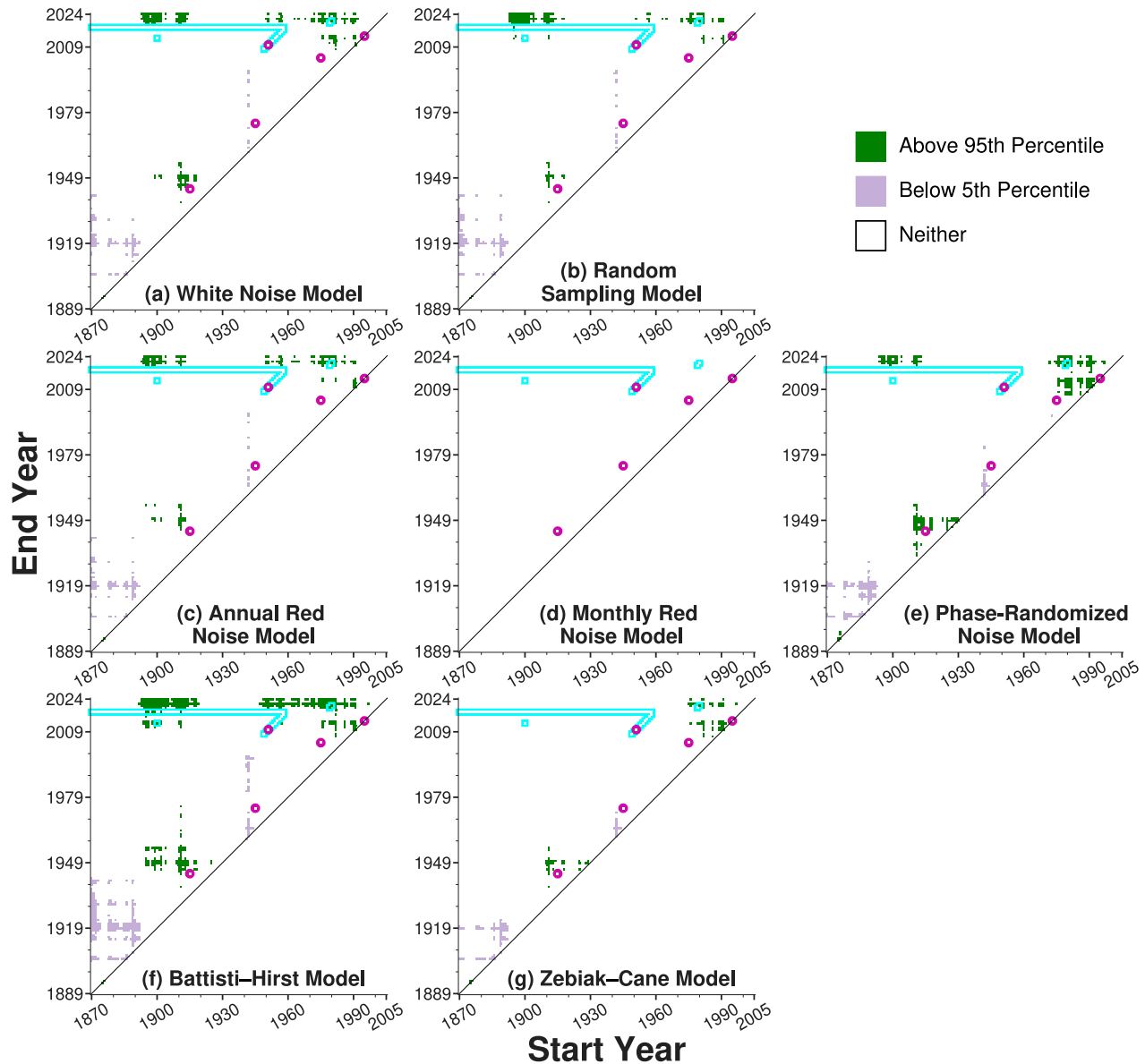


FIG. 10. As in Fig. 6, but using ERSST.v5 data.

SST datasets have been documented (e.g., Deser et al. 2010b), we perform the same analyses using four different interpolated SST datasets (HadISST, COBE, COBE2, and ERSST.v5), noting that uninterpolated datasets are not included due to their limited temporal coverage. Our results show that the key findings mentioned so far are broadly consistent across these four datasets (Figs. 7 and 11) and, hence, are likely not subject to observational uncertainties across the different SST datasets. Moreover, these findings highlight the importance of evaluating centennial-length trends, particularly to isolate the persistent, increasing influence of GHG forcing, given that anthropogenic GHGs are presumed to have dominated historical radiative forcings, especially in recent decades.

Notably, our findings are consistent with previous studies, regardless of whether they have argued that internal variability

or forced responses are the cause of the observed $\Delta\text{SST}_{\text{W-E}}$ trends. For instance, the $\Delta\text{SST}_{\text{W-E}}$ trends from the four intervals analyzed by Olonscheck et al. (2020) are estimated to be statistically insignificant across all the null hypothesis tests, though the intervals 1915–44, 1975–2004, and 1995–2014 are adjacent to significant trends for some tests and datasets (Figs. 6 and 8–11). This outcome suggests that these trends are likely due to internal variability, consistent with the conclusion of Olonscheck et al. (2020). Additionally, the 1951–2010 period assessed by Watanabe et al. (2021) and Seager et al. (2019) appears to be on the cusp between insignificant and significant trends, depending on the choice of null hypotheses and SST data products, which may explain why these two studies reached opposite conclusions. Similarly, for some null hypothesis tests and datasets, the region of the significant strengthening trends overlaps or is in

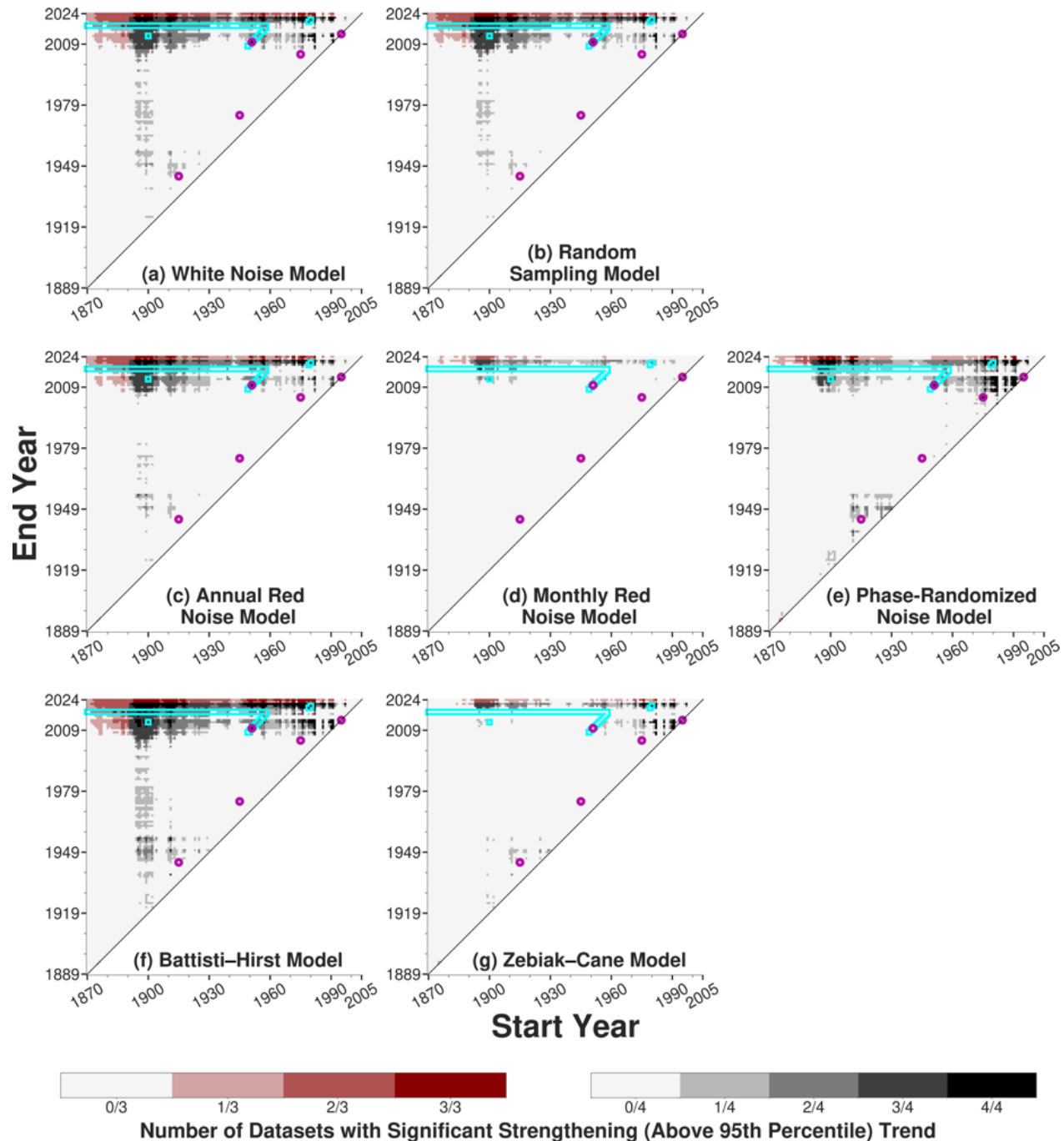


FIG. 11. Consistency in the statistical significance of observed $\Delta\text{SST}_{\text{W-E}}$ strengthening trends, demonstrated across HadISST, COBE, COBE2, and ERSST.v5. Each panel displays the summarized result of the hypothesis tests in the same order as in Figs. 6 and 8–10. Due to the limited temporal coverage of COBE, shades of red are used to indicate grids with a significant strengthening trend, starting before 1891 or ending after 2022; shades of gray are used otherwise. Cyan boxes and purple circles indicate the intervals used by past studies, as shown in Fig. 1a.

close proximity to the intervals studied in previous work that supported the argument that the observed trend is radiatively forced: 1979–2020 from Wills et al. (2022), 1980–2021 from Heede and Fedorov (2023), and the other nine 60-yr intervals between 1949 and 2017 from Seager et al. (2019). Over longer

intervals, such as 1900–2013 from Coats and Karnauskas (2017) and some centennial-scale intervals from Seager et al. (2022), our results indicate significance across many null hypothesis tests, except when tested against the two most conservative models (the monthly red noise model and Zebiak–Cane model)

or when the observed trends are estimated from ERSST.v5, further supporting the original conclusions of those studies. In short, these findings allow us to conclude that short-term trends are more likely to be subject to fluctuations due to internal variability, even though they may exhibit the most pronounced trends. This finding, in fact, aligns with a paleo-informed perspective that the recent 20-yr trend in the Walker circulation (i.e., zonal sea-level pressure gradient) from 1992 to 2011 sits within the reconstructed range of natural variability over the last 800 years (Falster et al. 2023). This again illustrates the importance of assessing centennial-length trends in $\Delta\text{SST}_{\text{W-E}}$ when evaluating human influences on tropical Pacific SSTs.

A caveat of this study is that the hypothesis tests considered may not be entirely realistic. For example, the white noise and random sampling models are conceptually least realistic due to their lack of SST persistence, though the estimated annual persistence of $\Delta\text{SST}_{\text{W-E}}$ appears to be very small in most datasets. The monthly red noise model is also unrealistic because it yields an annual persistence that is too high when compared to the estimate from the observed $\Delta\text{SST}_{\text{W-E}}$ (see Fig. 4). Based on the autocorrelation structure, we argue that the phase-randomized noise model appears to be the most realistic. The Battisti–Hirst and Zebiak–Cane models are also justifiable, as both are physically motivated and constructed. The Battisti–Hirst model also estimates an autocorrelation structure that is comparable to observations (Fig. 4), while the Zebiak–Cane model demonstrates a left-skewed distribution that aligns well with the observed distribution (Fig. 3g). However, the Battisti–Hirst model exhibits a bimodal distribution (Fig. 3f), while the Zebiak–Cane model has interannual cycles that are too regular compared to observations, the consequence of which is an unrealistic autocorrelation structure (see Fig. 4). Furthermore, extratropical–tropical and interbasin teleconnections are not accounted for in either the Battisti–Hirst or Zebiak–Cane model, while they may have been encoded in the red noise models and the phase-randomized noise model that we construct based on observed $\Delta\text{SST}_{\text{W-E}}$ time series, as such processes are inherent in observations. Regardless of which null hypothesis is the most realistic, our results across all significance tests favor our key finding that significant trends over intervals with recent end dates are mostly strengthening, and thus, these trends, especially the longer ones, may be better explained by a response to historical radiative forcings on top of internal variability.

The observed $\Delta\text{SST}_{\text{W-E}}$ trends may reflect the combined effects of internal variability and historical radiative forcings, including GHGs, aerosols, and stratospheric ozone, depending on the timing of the trend intervals. While our findings indicate a potential role of these forcings in driving the observed strengthening trends over intervals with recent end dates, pinpointing specific forcing agents is beyond the scope of this study. Nonetheless, the unprecedented increase in atmospheric GHG concentrations likely dominates these historical radiative forcings (e.g., Arias et al. 2021), especially in recent decades. This has led us to speculate that anthropogenic GHGs play a particularly prominent role in shaping the observed, statistically significant, recent strengthening trends. We also cautiously note that while some studies argue that the tropical

Pacific Ocean will eventually show a weakening of $\Delta\text{SST}_{\text{W-E}}$ based on the consistency with paleoclimatological evidence (e.g., Tierney et al. 2019; Watanabe et al. 2024), the implications of our analyses and results are relevant to the historical and the near-term future transient responses. Furthermore, this study is not a dynamical assessment of the $\Delta\text{SST}_{\text{W-E}}$ trends. We, therefore, do not identify the atmospheric and oceanic processes that govern the observed trends or elucidate the cause of the discrepancies between observations and climate models. Myriad causes have been proposed (Watanabe et al. 2024), including model misrepresentation of processes in the equatorial Pacific Ocean (e.g., Coats and Karnauskas 2017; Seager et al. 2019) and the remote response to SST changes in other tropical basins (e.g., McGregor et al. 2018) and the Southern Ocean (e.g., Kang et al. 2023; Kim et al. 2022), some of which may be related to models' resolution-dependent ability to resolve mesoscale ocean eddies (e.g., Kang et al. 2026). Aerosol forcing could also have contributed to the observed $\Delta\text{SST}_{\text{W-E}}$ trends (e.g., Heede and Fedorov 2021) and might be subject to uncertainties in emissions and modeled interactions with clouds and radiation (e.g., Fiedler et al. 2023; Ying and Huang 2016; Zelinka et al. 2014). Recently, Jiang et al. (2024a,b, 2025) have additionally drawn attention to the role of subsurface cooling, thermocline shoaling, and their connection through upwelling and mixing into the mixed layer, which are distinct between observations and model simulations. In particular, they identify a spatial pattern of winds, SST, and upper-ocean circulation and thermal structure in observations that is dynamically distinct from that due to the natural decadal variability of the PDO/IPO. They argue that this Pacific Climate Change (PCC) pattern is emerging over time and is consistent with a forced response. Our findings are in agreement with these studies, as they demonstrate that trends with recent end dates cannot be explained in terms of internal variability, which we assessed using null hypothesis tests based on multiple statistical and physically based simple models. Future work should compare all the possible observed trends with climate model ensembles (e.g., Byrne et al. 2025). Particular interest should be paid to whether the observed strengthening trends ending in the recent period, especially those of centennial length, can be reproduced by the models as a result of internal variability alone or some combination of variability and a response to historical radiative forcings, particularly anthropogenic GHG forcing. Nonetheless, our comprehensive assessment suggests that observed strengthening $\Delta\text{SST}_{\text{W-E}}$ trends are a robust feature of the equatorial Pacific SSTs and that those ending in recent years, being beyond the range of a variety of statistical, conceptual, and geophysical models, are likely a radiatively forced, transient response of the tropical Pacific Ocean to historical forcings, primarily anthropogenic GHG forcing.

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Data availability statement. All four observational SST records are publicly available: HadISST through <https://www.metoffice.gov.uk/hadobs/hadisst/>, COBE through <https://downloads.psl.noaa.gov/Datasets/COBE/>, COBE2 through <https://downloads.psl.noaa.gov/Datasets/COBE2/>, and ERSST.v5 through <https://iridl.ldeo.columbia.edu/SOURCES/NOAA/NCDC/ERSST/version5/sst/>.

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